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Shripad Vilasrao Deshmukh

Current Research Focus

Evaluation-aware decision making; Non-convex policy learning; RL finetuning of language models

Education

Fall 2023 – **Integrated MS/PhD in Computer Science (Year IV)**, *University of Massachusetts*
GPA: 3.97/4.0, Advisor: Prof. Scott Niekum

2016 – 2020 **BTech in Electrical Engineering**, *Indian Institute of Technology Madras*
CGPA: 9.20/10

Publications (Google Scholar Profile)

Preprints and Drafts

2026 Y Chittepu*, D Gupta*, **S Deshmukh***, P Thomas, S Niekum. *TRIBE: Trust-Region Policy Improvement from Batch Experience*. Under submission. (*Equal contribution)

2026 **S Deshmukh**, Y Chittepu, D Gupta, P Thomas, S Niekum. *Convex-Concave Reinforcement Learning*. Under submission.

2026 **S Deshmukh**, C Bock, K Ruan, A Awasthi, V Lingam, R Houthooft, M Prajapat. *Projected Distillation: On learning to solve hard problems using close, correct targets*. NeurIPS 2026 Workshop on Test-Time Continual Learning. Under submission elsewhere.

2026 K Kim, **S Deshmukh**, N Vlassis, J Zhang. *Inverse Reinforcement Learning without an Optimal Demonstrator: A Feasible Reward Set Approach*. Under submission.

2025 **S Deshmukh**, W Schwarzer, S Niekum. *Evaluation-Aware Reinforcement Learning*. Under submission.

Conference Publications

2026 **S Deshmukh**, J Subramanian, R Addanki, N Vlassis. *COSAC: Counterfactual Credit Assignment in Sequential Cooperative Teams*. Neural Information Processing Systems (NeurIPS) 2026.

2025 S Dandekar, **S Deshmukh**, F Chiu, W B Knox, S Niekum. *A Descriptive and Normative Theory of Human Beliefs in RLHF*. Transactions on Machine Learning Research (TMLR) 2025.

2023 S Singh, **S Deshmukh**, M Sarkar, B Krishnamurthy. *LOCATE: Self-supervised Object Discovery*. British Machine Vision Conference (BMVC) 2023.

2023 **S Deshmukh**, A Dasgupta, C Agarwal, N Jiang, B Krishnamurthy, G Theocharous, J Subramanian. *Explaining RL Decisions with Trajectories*. International Conference on Learning Representations (ICLR) 2023.

2023 A Java*, **S Deshmukh***, M Aggarwal, S Jandial, M Sarkar, B Krishnamurthy. *One-Shot Doc Snippet Detection: Powering Search in Document Beyond Text*. IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2023. (*Equal contribution)

2020 P Gupta, N Puri, S Verma, D Kayastha, **S Deshmukh**, B Krishnamurthy, S Singh. *Explain Your Move: Understanding Agent Actions Using Focused Feature Saliency*. International Conference on Learning Representations (ICLR) 2020.

Patents (with Attorney Docket Numbers)

1. *Critic-Free Counterfactual Advantage Estimation for Sequential MAS*, P29615-US-1†
2. *Novel Self-supervised Object Discovery in Videos*, P12351-US
3. *Account Executive Actionable Digest*, P12314-US†
4. *Relational Loss for Enhancing Text-based Style Transfer*, P12300-US
5. *A Framework for Leveraging LLM Models and RL in Marketing Decision Making*, P12223-US†
6. *Video Object Segmentation through Flow-guided Graph-cut*, P12170-US
7. *Novel method to propagate personalized error corrections across Forms corpus*, P12004-US†
8. *Novel Trajectory-based Explainability Framework for RL-based Decision Making*, P11853-US†
9. *Forms Similarity Matching Framework for Enhancing RnC tool in AEM Forms*, P11882-US†
10. *A Novel Multimodal One-Shot Detection Approach for Document Snippet Search*, P11686-US†
11. *Semantic Noise based Soft Label Regularization for Distilling Model Knowledge*, P11539-US
12. *Novel method to simplify data points for easier understanding of neural networks*, P11364-US†
13. *Novel Method and Apparatus to Control Diffusion Model Image Generation*, P11343-US
14. *Refining Element Associations for Form Structure Extraction*, P10768-US†
15. *Customer Journey Management Using Machine Learning*, P10405-US†

†First inventor or significant contributor. See google scholar profile for details.

Research & Industry Experience

Research Intern, *Amazon Kiro Science*, Santa Clara, Jun'26 – Sept'26

- Developed a distillation technique for training small LLMs to specialize on hard tasks. Our approach uses minimal edits to the wrong tokens of the small LLM's response for correction.
- Produced strong empirical results across diverse task domains and insightful theoretical guarantees for learning regret. Work accepted at NeurIPS'26 TTCL Workshop and under submission at a prominent ML conference.

Research Intern, *Adobe Research*, San Jose, May'25 – Aug'25

- Developed policy gradient algorithms for multi-agent large language model systems that achieve faster convergence via variance reduction in the per-agent policy gradient. Work accepted at NeurIPS'26. Filed a US patent on the method.

Machine Learning Research Associate, *Adobe India*, Noida, Feb'22 – Jul'23

- Proposed counterfactual-based RL explainability framework. Published at ICML'23 workshop on counterfactuals.
- Proposed a data-grounded explainability approach for offline RL – attributing agent's current decisions to the past trajectory experiences. Published at ICLR'23.
- Designed and implemented transformer-based document processing framework, MONOMER, outperforming state-of-the-art document analysis and one-shot detection approaches. Work published at WACV'23.
- Filed 12 US patents in the areas of marketing decision making and multimodal learning.
- *Noteworthy implementations*: (i) Decision transformers with RL-centric losses, (ii) LLMs for marketing decision making, (iii) Decision RNNs, (iv) Deep agent-based models.

Machine Learning Engineer, *Adobe India*, Work from Home, Jun'21 – Jan'22

- Maintainer of Adobe's automated form conversion service. Rigorously converted Adobe's internal form corpus to standard COCO format for streamlined training and evaluation. This led to state-of-the-art conversion quality.

Member of Technical Staff, *Adobe India*, Work from Home, Aug'20 – May'21

- Enhanced form element association using segmentation masks, line text and spatial characteristics (15% precision and 4% recall gain). Drafted patent-idea, 'Refining form structures. . . ', now listed as the *exemplary patent draft* on Adobe Brightidea.
- Proposed novel trajectory-ranking based inverse RL for reducing customer churn rate in marketing. Patented the idea.

Summer Research Intern, *Adobe India*, Noida, May'19 – Aug'19

- Co-developed a new saliency metric for visual explanations of RL agents and applied it to Chess engines (Stockfish, LeelaChess-0), Go (minigo), and Atari agents. Published at ICLR'20.

R&D Intern, *Center for Development of Advanced Computing (CDAC)*, Pune, Dec'18 – Jan'19

- Developed FORTRAN libraries as part of the National Supercomputing Mission project.

Relevant Graduate-level Coursework

CS690S: Human-Centric Machine Learning · CS690F: Responsible AI · CS603: Robotics · CS682: Neural Networks · CS6700: Reinforcement Learning · EE6418: Game Theory · EE6132: Deep Learning for Computer Vision · EE5121: Convex Optimization

Awards & Achievements

2017 National Prize Certificate for top students at IIT Madras (Top 7%)

2016 All India Rank 323 in JEE Advanced (Top 0.16%) and AIR 491 in JEE Mains (Top 0.03%)

2015 Ranked among top 1% in National Chemistry Olympiad, India

2015 KVPY Fellowship

2012 NTSE Scholarship

Academic Service & Mentorship

Reviewer NeurIPS'23, ICLR'24, ICLR'25, ICML'25, AAAI'25, ICLR'26, NeurIPS'26 (**Top Reviewer**). OpenReview profile

App Reviewer Assessed 60+ candidate applications for RL PhD positions at UMass for 2025 admission cycle.

Mentor Mentored 3 undergrads exploring transcendence in AI (Summer 2025) and 9 undergrad research interns on topics in RL and multimodal learning at Adobe (2021-2023).

Leadership & Outreach

Organizer Organized Adobe Digital Experience (DX) ML Hackathon (2023).

Moderator Led Liberty, Equality & Fraternity (LEAF) society at IIT Madras, facilitating discussions on Indian socio-economic issues and technological solutions (2020).

Organizer Organized principal lectures at IIT Madras featuring luminaries including Orkut Büyükkökten, Nirmala Sitharaman, and Dr. Prakash Amte. Managed the proceedings of off-campus lecture by Prof. Babu Vishwanathan (2017-18).